Use the Bisection method to find p_3 for $f(x) = \sqrt{x} - \cos x$ on $[0, 1]$.

n	a_n	b_n	p_n	$f(p_n)$	ε_a
1	0	1	0.5	-0.170	—
2	0.5	1	0.75	0.134	2.269
3	0.5	0.75	0.625	-0.0204	1.152

Use the Bisection method to find solutions accurate to within 10^{-5} for the following problems.

a. $x - 2^{-x} = 0$ for $0 \le x \le 1$

n	a_n	b_n	p_n	$f(p_n)$	ε_a
1	0	1	0.5	-0.207	—
2	0.5	1	0.75	0.155	0.333
3	0.5	0.75	0.625	-0.0234	0.2
4	0.625	0.75	0.6875	0.0666	0.091
5	0.625	0.6875	0.65625	0.0217	0.0476
6	0.625	0.65625	0.640625	-0.00081	0.0244
7	0.640625	0.65625	0.6484375	0.0105	0.0120
8	0.640625	0.6484375	0.64453125	0.00483	0.0061
9	0.640625	0.64453125	0.642578125	0.0020	0.0030
10	0.640625	0.642578125	0.641601562	0.0006	0.00152
11	0.640625	0.641601562	0.641113281	-0.000104	0.000761
12	0.641113281	0.641601562	0.641357422	0.000247972	0.000381
13	0.641113281	0.641357422	0.641235352	0.000071651	0.000190
14	0.641113281	0.641235352	0.641174316	-0.000016	0.0000952
15	0.641174316	0.641235352	0.641204834	0.000027573	0.0000475
16	0.641174316	0.641204834	0.641189575	0.000005533	0.0000238
17	0.641174316	0.641189575	0.641181946	-0.000002489	0.0000119

Notice also that $|p - p_n| < \frac{|b-a|}{2^n} = \frac{1}{2^n} < 10^{-5}$

$2^n > 10^5 \Rightarrow n > \frac{5}{\log_{10} 2} = 16.61 \Rightarrow \boxed{n \ge 17}$

Find an approximation to $\sqrt{3}$ correct to within 10^{-4} using the Bisection Algorithm.

Consider the function $f(x) = x^2 - 3$. Thus, we want to find the root of this function. Since $f(1) = -2$, $f(2) = 1$ $f(p) = 0$ with $p \in [-2, 1]$

n	a_n	b_n	P_n	$f(P_n)$	ε_a
1	1	2	1.5	-0.75	—
2	1.5	2	1.75	0.0625	0.167
3	1.5	1.75	1.625	-0.359	0.077
4	1.625	1.75	1.6875	-0.1523	0.037
5	1.6875	1.75	1.71875	-0.0459	0.0182
6	1.71875	1.75	1.734375	0.0081	0.0090
7	1.71875	1.734375	1.7250625	-0.02415	0.0054
8					
9		and	so		
10			on		
11					
12					
13					
14					

Notice

$|P - P_n| < \frac{1}{2^n} < 10^{-4}$

$\Rightarrow n > \frac{4}{\log(2)} > 13.2$

$\Rightarrow n \geq 14.$

Doing this on Mathematica gives

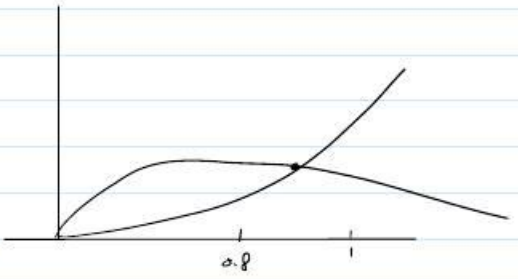
$P_{14} = 1.73206$

5.5 Locate the first nontrivial root of $\sin x = x^3$ where x is in radians. Use a graphical technique and bisection with the initial interval from 0.5 to 1. Perform the computation until ε_a is less than $\varepsilon_s = 2\%$. Also perform an error check by substituting your final answer into the original equation.

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$f(x) = \sin x - x^3$

graphically



$p \in [0.8, 1]$

$f(0.8) = 0.205 > 0$

$f(1) = -0.1585 < 0$

n	a_n	b_n	P_n	$f(P_n)$	ε_a
1	1	0.8	0.9	0.054	—
2	1	0.9	0.95	-0.043	0.052
3	0.95	0.9	0.925	0.007	0.027
4	0.95	0.925	0.9375	-0.017	0.013

Use the method of False Position to find a solution to $x = \cos x$

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$$P_n = P_{n-1} - \frac{f(P_{n-1})\,(P_{n-1}-P_{n-2})}{f(P_{n-1}) - f(P_{n-2})} \qquad \text{with } \varepsilon_a < 10^{-4}$$

$f(x) = \cos x - x$, $x \in [0.5,\ \pi/4]$

$f(0.5) = 0.377 > 0$

$f(\pi/4) = -0.078 < 0$

n	a_n	b_n	P_n	$f(P_n)$	ε_a
0	0.5	$\pi/4$	0.736841 3884	0.00451771852	-
1	0.736841 3884	$\pi/4$	0.739058 13929	0.0000451771709	0.003618
2	0.739058 13929	$\pi/4$	0.739084 86382	0.0000004 5087	3.6×10^{-5}
3					

The fourth-degree polynomial

$$f(x) = 230x^4 + 18x^3 + 9x^2 - 221x - 9$$

has two real zeros, one in $[-1, 0]$ and the other in $[0, 1]$. Attempt to approximate these zeros to within 10^{-6} using the

a. Method of False Position

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$$P_n = P_{n-1} - \frac{f(P_{n-1})\,(P_{n-1}-P_{n-2})}{f(P_{n-1}) - f(P_{n-2})}$$

we will only make the zero $\in [0,1]$ $f(0) = -9 < 0$, $f(1) = 27 > 0$

n	P_{n-2}	P_{n-1}	P_n	$f(P_n)$	ε_a
2	0	1	0.25	-62.5078	-
3	1	0.25	0.773763	-62.6058	0.703
4	1	0.773763	0.95219	-6.66427	0.116
5	1	0.95219	0.961655	-0.494843	9.8×10^{-3}
6	1	0.961655	0.962345	-0.0035606	7.1×10^{-4}
7	1	0.962345	0.962395	-0.00256	5.2×10^{-5}
8	1	0.962395	0.962398	-0.000164	3.1×10^{-6}
9	1	0.962398	0.9623984	-2×10^{-5}	4.1×10^{-7}